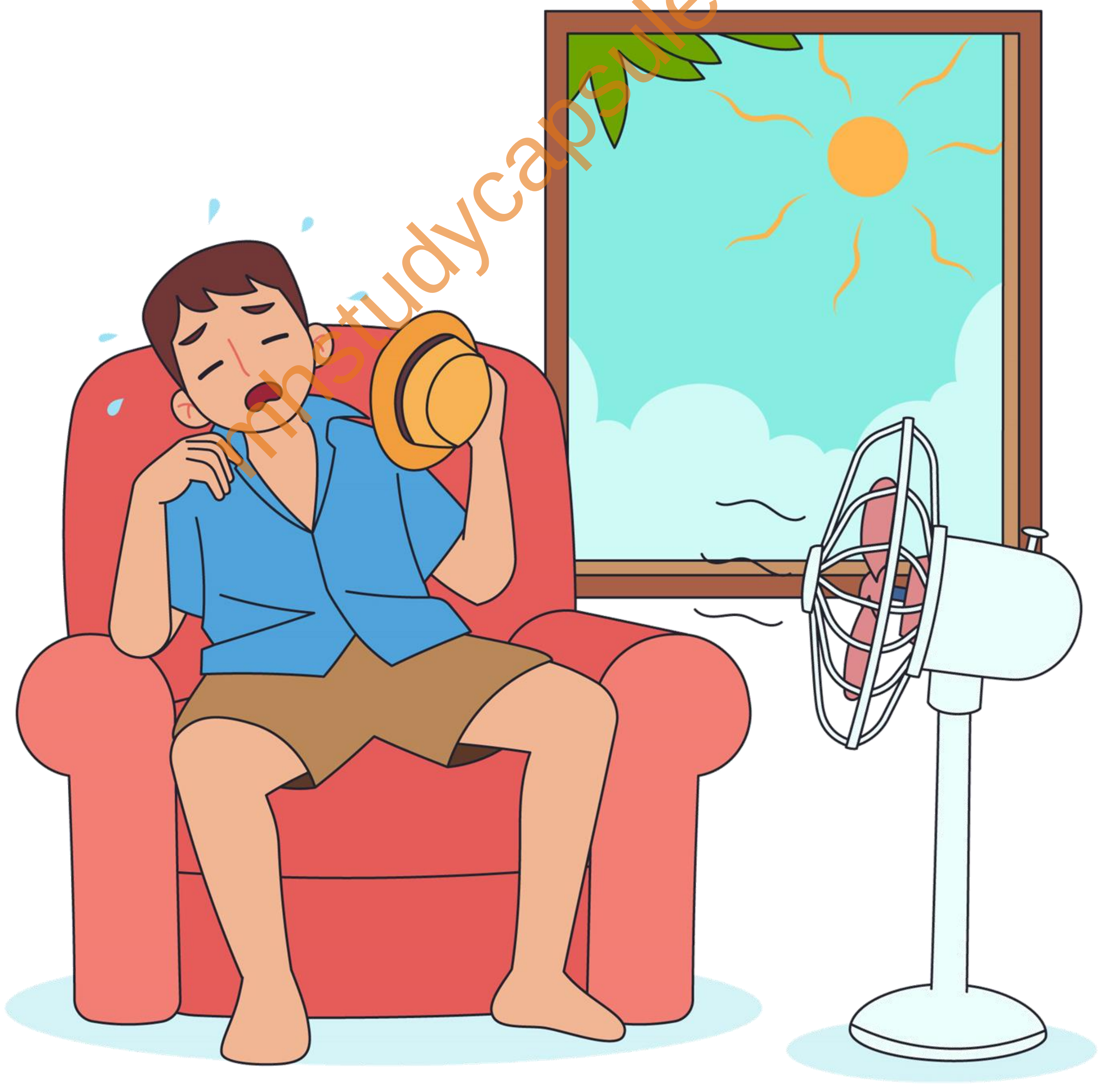
Chapter 5

HEAT



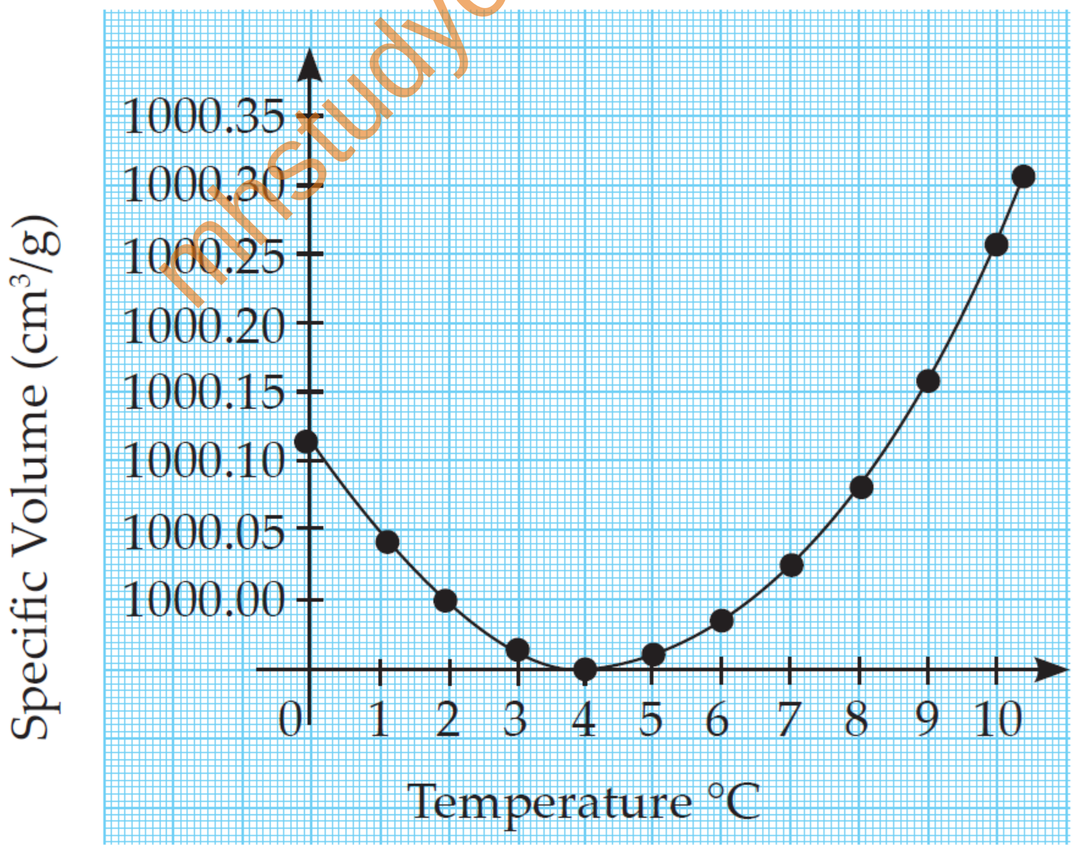
1. **Fill in the blanks and rewrite the sentence.**

a. The amount of water vapour in air is determined in terms of its **Absolute humidity.**

b. If objects of equal masses are given equal heat, their final temperature will be different. This is due to difference in their **specific heat capacities.**

c. During transformation of liquid phase to solid phase, the latent heat is **given out (released).**

2. **Observe the following graph. Considering the change in volume of water as its temperature is raised from 0ºC, discuss the difference in the behaviour of water and other substances. What is this behaviour of water called?**



i. Most substances expand on heating and contract on colling.

ii. Whereas, from the graph, it is clearly visible that water shows a distinct and peculiar behaviour between $0^{o}C$ to $4^{o}C$.

iii. Water, instead of expanding, contracts between $0^{o}C$ to $4^{o}C$.

iv. After $4^{o}C$, it shows the normal behaviour of expansion as is shown by other substances.

v. Thus, at $4^{o}C$, water possesses maximum density and minimum volume. This behaviour of water between $0^{o}C$ to $4^{o}C$ is known as anomalous behaviour of water.

3. What is meant by specific heat capacity? How will you prove experimentally that different substances have different specific heat capacities?

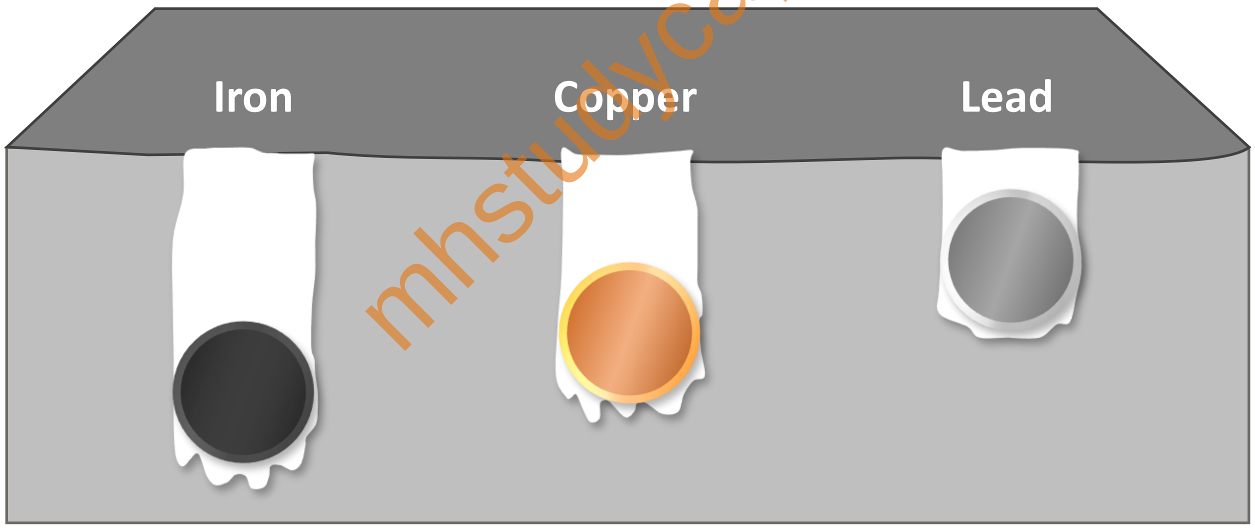


Image just for reference. Not to be drawn.

i. The quantity of heat energy required to raise a temperature of unit mass of a substance through 1°C is called specific heat capacity of a substance.

ii. Three solid spheres of equal masses of material iron, copper and lead are placed in boiling water. After removing them simultaneously, they are immediately placed on a wax slab.

iii. Initially, all spheres have temperature of 100°C. The sphere which absorbs more heat can melt more wax.

iv. Iron sphere melts more wax and goes deep into it ray. The lead sphere goes less deep into the molten wax. The copper sphere goes to intermediate depth.

v. This shows that for equal rise in temperature, the three spheres have absorbed different amounts of heat.

vi. This means that the property which determines the amount of heat absorbed by a sphere is different for the three spheres.

vii. This property is called the specific heat capacity.

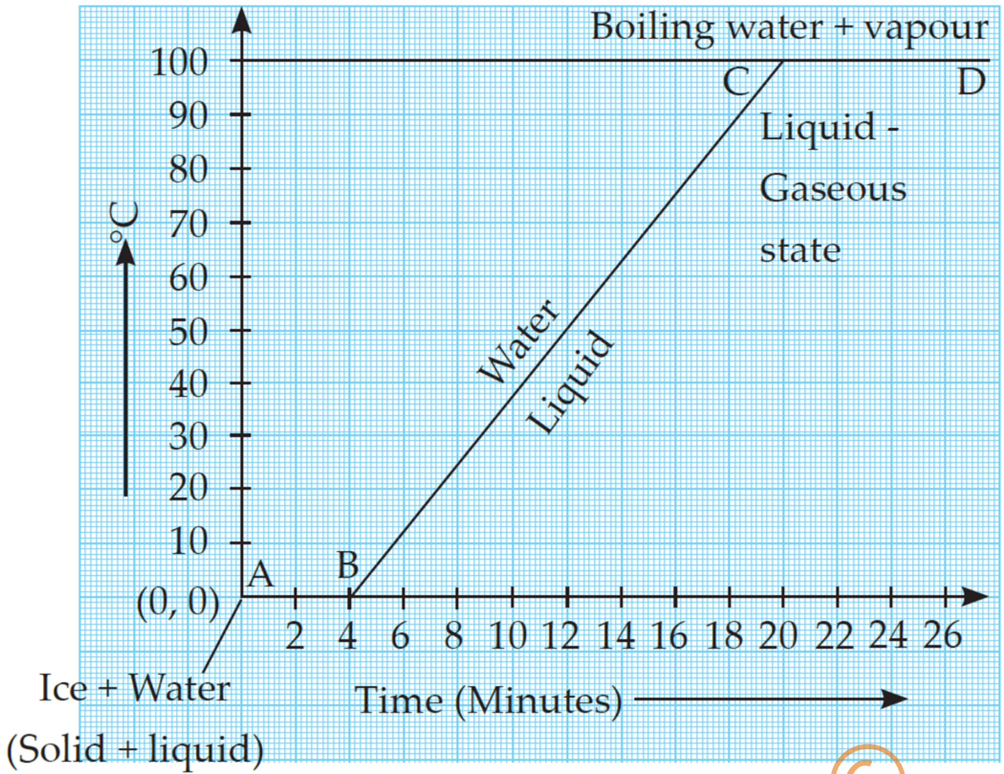
viii. Specific heat capacity is denoted by letter 'c'. In CGS, it is measured in cal/g°C, while in MKS, it is measured in J/kg°C.

4. While deciding the unit for heat, which temperatures interval is chosen? Why?

i. The temperature range of 14.5°C to 15.5°C is chosen while deciding the unit for heat.

ii. Because, if we heat 1 kg of water by 1°C in different temperature range other than 14.5°C to 15.5°C, the amount of heat required will be slightly different.

5. Explain the following temperature versus time graph.



i. The graph shown is a temperature - time graph of Ice transforming into water and steam.

ii. AB shows that, at constant temperature of 0°C ice changes to water.

iii. The temperature at which ice changes to water, is called melting point of ice i.e., 0°C.

iv. Now, water at 0°C starts heating and its temperature gradually increases to 100°C.

v. BC shows increase in temperature from 0°C to 100°C without a change in state.

vi. CD shows that, at constant temperature of 100°C, water changes to vapour.

vii. The temperature at which water changes to vapour is called boiling point of water i.e., 100°C.

6. Explain the following:

a. What is the role of anomalous behaviour of water in preserving aquatic life in regions of cold climate?

i. Water shows anomalous behaviour. It contracts from 0°C to 4°C and beyond 4°C it expands. So, the density of water is maximum at 4°C.

ii. However, when the surrounding temperature goes down, the water in oceans and rivers also cools down and the temperature of water reaches 4°C.

iii. The water therefore reaches its maximum density at this temperature. Below this temperature (4°C), the water layer on the surface expands due to anomalous behaviour of water (because of decrease in density).

iv. Hence, this colder layer remains on top and converts into ice. This ice acts as an insulator and does not allow the temperature of the water layer below it to fall below 4°C.

v. This ensures that a livable temperature is maintained for aquatic life under the oceans and rivers.

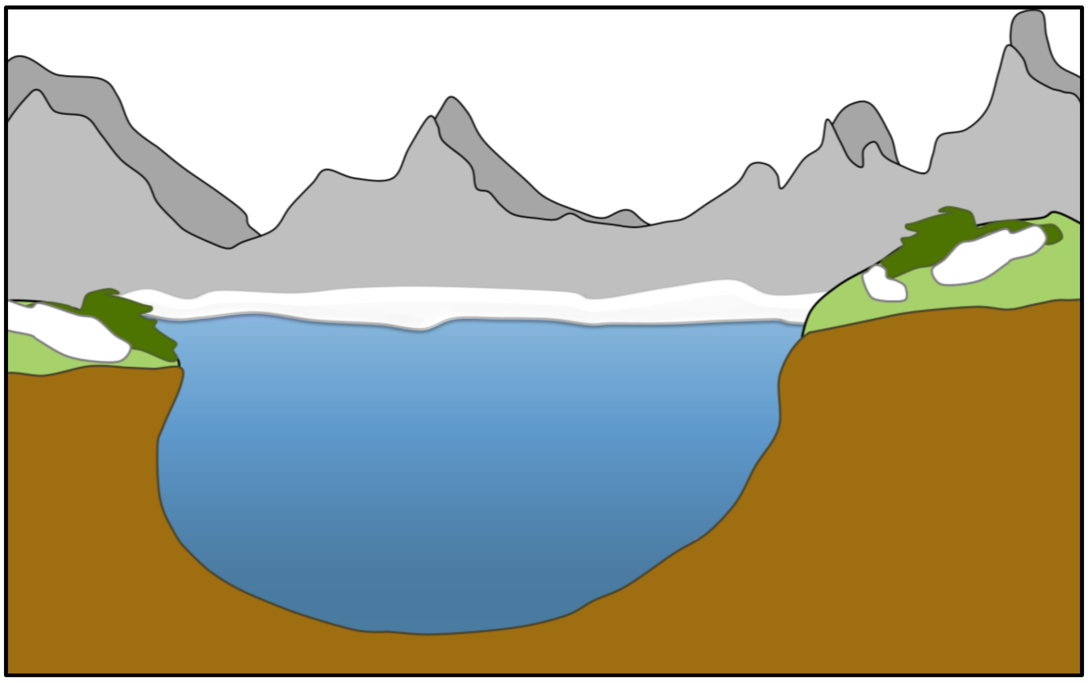


Image just for reference. Not to be drawn.

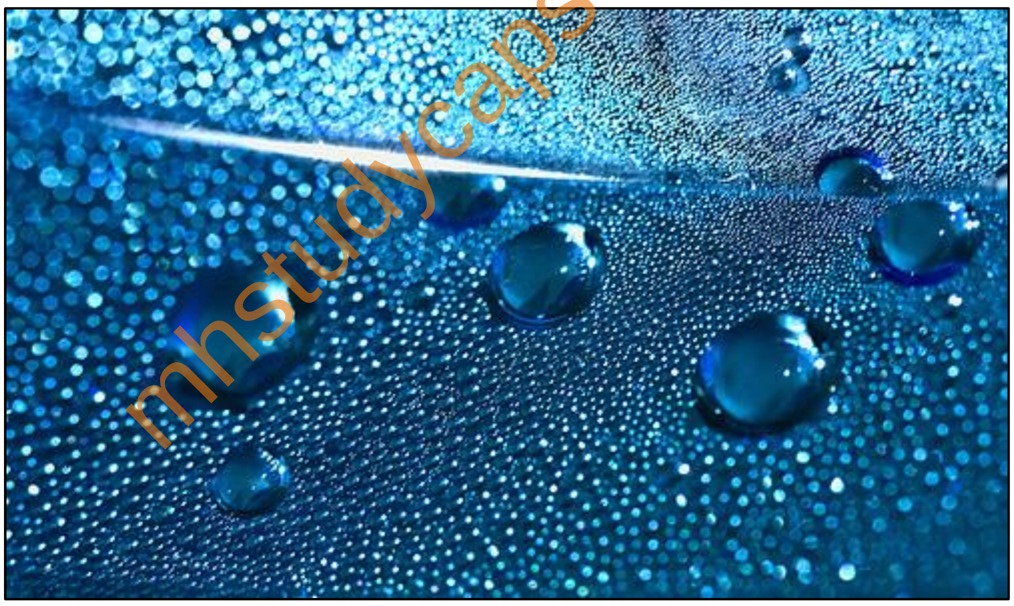
b. How can you relate the formation of water droplets on the outer surface of a bottle taken out of refrigerator with formation of dew?

i. When the bottle is taken out of the refrigerator, the air around the bottle cools.

ii. As the air cools, due to decrease in temperature it becomes saturated with water vapour.

iii. As a result, the excess water vapour gets converted into tiny droplets on the outer surface of the bottle called dew.

iv. This is like the dew we see in early mornings on leaves of plants and window glass of vehicle.



c. In cold regions in winter, the rocks crack due to anomalous expansion of water.

i. During winter season, in cold countries the temperature of atmosphere falls well below 0°C.

ii. If water is present within the crevices of the rocks, then due to its anomalous behaviour, water begins to expand below 4°C.

iii. As there is no place in the crevices for the expansion of water, it exerts tremendous pressure on the rocks which results in the crumbling of the rocks.

iv. Hence, sometimes rocks crumble into pieces in cold countries during winter.

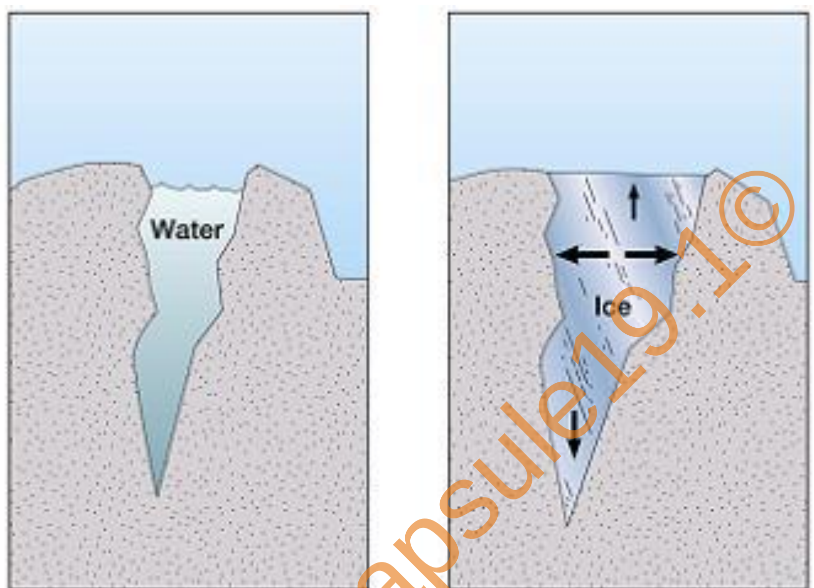


Image just for reference. Not to be drawn.

7. Answer the following:

a. What is meant by latent heat? How will the state of matter transform if latent heat is given off?

i. The quantity of heat absorbed or released by a substance to change its state at constant temperature is called latent heat.

ii. If Latent heat is given off, then a gas will convert into a liquid, and a liquid will convert into a solid.

b. Which principle is used to measure the specific heat capacity of a substance?

i. Principle of heat exchange is used in the calorimetry method to determine the specific heat capacity of a substance.

ii. It states that, Heat lost by hot object = Heat gained by cold object.

c. Explain the role of latent heat in the change of state of a substances?

i. The heat energy absorbed at constant temperature during transformation of solid into liquid is called the latent heat of fusion.

ii. This heat energy is utilized for weakening the bonds between the atoms or molecules in the solid and transform it into liquid phase.

iii. Similarly, the heat energy absorbed at constant temperature during transformation of liquid into gas is called the latent heat of vapourization.

iv. This heat energy is utilized for weakening the bonds between the atoms or molecules in the liquid and transform it into gaseous phase.

v. During transformation of gaseous phase to liquid phase and from liquid phase to solid phase, latent heat is given out by the substance to the atmosphere.

d. On what basis and how will you determine whether air is saturated with vapour or not?

i. We can determine whether air is saturated with vapour or not on the basis of relative humidity.

ii. Relative humidity is the ratio of actual mass of vapour content in air for a given volume and temperature to that required to make air saturated with vapour at that temperature.

iii. If relative humidity is 100%, the air is saturated with water vapour. This is called dew point.

iv. If the value is less than 100%, then the air is unsaturated.

v. If the relative humidity is more than 60%, we feel that the air is humid. If the relative humidity is less than 60%, we feel that the air is dry.

8. Read the following paragraph and answer the questions.

If heat is exchanged between a hot and cold object, the temperature of the cold object goes on increasing due to gain of energy and the temperature of the hot object goes on decreasing due to loss of energy. The change in temperature continues till the temperatures of both the objects attain the same value. In this process, the cold object gains heat energy and the hot object loses heat energy. If the system of both the objects is isolated from the environment by keeping it inside a heat resistant box (meaning that the energy exchange takes place between the two objects only), then no energy can flow from inside the box or come into the box.

i. **Heat is transferred from where to where?**

Heat gets transferred from hot object to the cold object.

ii. **Which principle do we learn about from this process?**

Principle of Heat exchange.

iii. **How will you state the principle briefly?**

The principle of Heat exchange states that, if heat is exchanged between a hot and cold object, then heat energy lost by hot object is equal to heat energy gained by the cold object.

iv. **Which property of the substance is measured using this principle.**

Using this principle, the specific heat capacity of a substance can be measured.

INTEXT QUESTIONS

1. What is the difference between heat and temperature?

Heat	Temperature
It is a form of energy.	It is a measure of degree of hotness or coldness of an object.
SI unit is joule (J).	SI unit is kelvin (K).
Two bodies of different materials having same quantity of heat can be at different temperatures.	Two bodies at same temperature can have different quantities of heat.
It can be measured by using Calorimeter.	It can be measured using Thermometer.

2. What are the different ways of heat transfer?

Heat can be transferred in three ways:

i. Conduction

ii. Convection

iii. Radiation

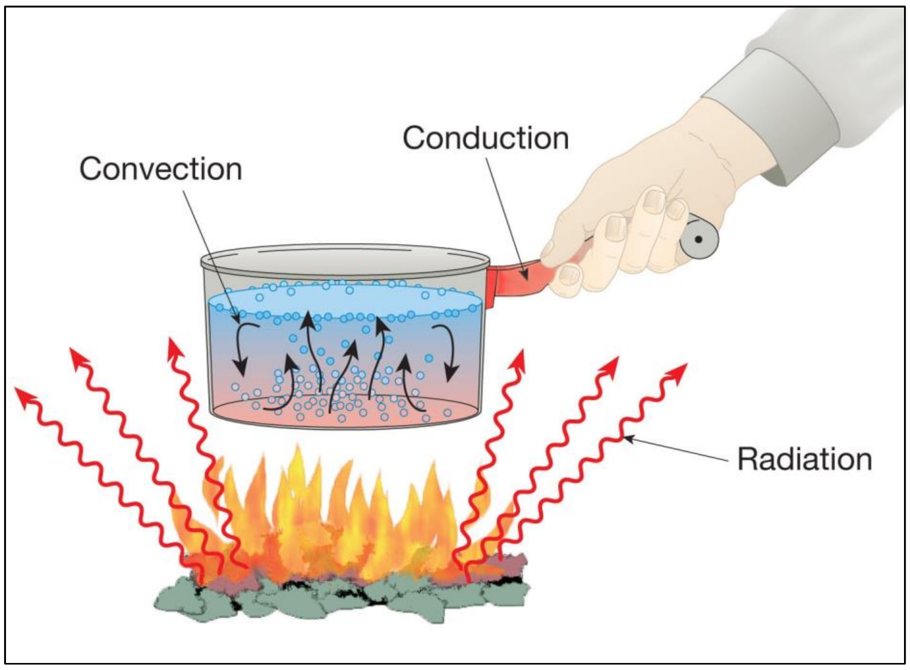


Image just for reference. Not to be drawn.

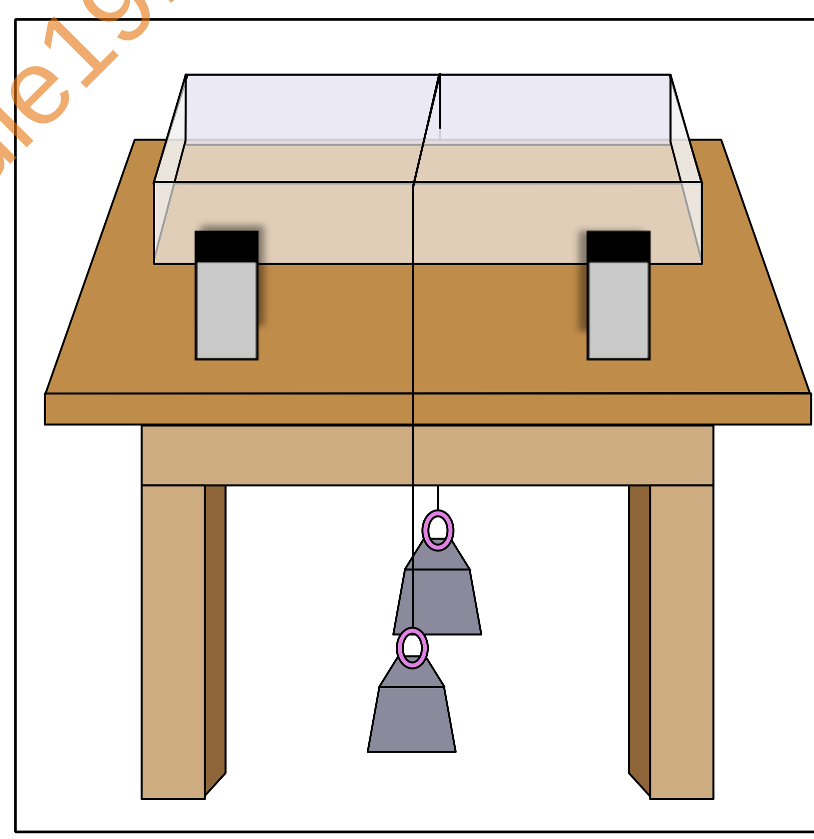
3. **Is the concept of latent heat applicable during transformation of gaseous phase to liquid phase and from liquid phase to solid phase?**

Yes, the concept of Latent heat is applicable in both the cases. The heat energy will be evolved (given out) from the substance.

4. **Where does the latent heat go during these transformations?**

The latent heat given out during these transformations is released into the atmosphere.

5. **In this experiment, the wire moves through the ice slab. However, the ice slab does not break. Why?**



i. In the experiment it is observed that the wire gradually penetrates the ice slab.

ii. After some time, the wire comes out of the lower surface of the ice slab. However, the ice slab does not break.

iii. The melting point of ice becomes lower than $0^{o}C$ due to pressure. This means that at $0^{o}C$, the ice gets converted into water.

iv. As soon as the pressure is removed, the melting point is restored to $0^{o}C$ and water gets converted into ice again.

v. The phenomenon in which the ice converts to liquid due to applied pressure and then re-converts to ice once the pressure is removed is called regelation.

6. Is there any relationship of latent heat with the regelation?

i. Yes, there is a relationship between latent heat and regelation.

ii. The amount of heat energy absorbed or released during change of state is called Latent heat.

iii. The phenomenon in which ice converts to liquid due to applied pressure and then reconverts to ice once the pressure is removed is called Regelation.

iv. The relationship to latent heat is that when ice melts under pressure, it absorbs heat due to the latent heat of fusion, and when it refreezes, it releases the absorbed heat.

7. You know that as we go higher than the sea level, the boiling point of water decreases What would be effect on the melting point of solid?

i. The melting point of a solid is primarily determined by the substance's intrinsic properties and the strength of the chemical bonds that hold its particles together.

ii. So, the melting point of a solid remains relatively constant as you change altitude or atmospheric pressure, unlike the boiling point of liquids, which can be significantly affected by pressure changes.

8. We feel that some objects are cold, and some are hot. Is this feeling related in some way to our body temperature?

i. If two objects are in contact with each other, there will be an exchange of heat between them.

ii. We feel that an object is hot, when our body temperature is low as compared to that of the object because heat is transferred from object to body.

iii. Similarly, when body temperature is high as compared to that of object, we feel that the object is cold because heat is transferred from body to object.

9. In regions with cold climate, the aquatic plants and animals can survive even when the atmospheric temperature goes below 0^oC.

– Exercise Question 6 - a

10. In cold regions in winter the pipes for water supply break.

i. During winter season, in cold countries the temperature of atmosphere falls well below 0°C.

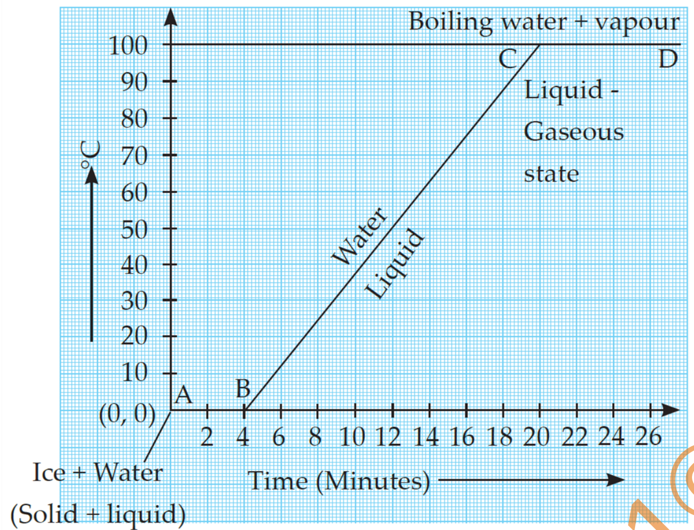
ii. Therefore, the temperature of water inside the pipeline also decreases.

iii. When the temperature of the water falls below 4°C, it starts expanding instead of contracting due to anomalous behaviour of water.

iv. Since there is no place for water to expand it exerts tremendous pressure on the walls of the pipeline causing them to burst. Hence, in winter the pipeline carrying water burst in cold countries.

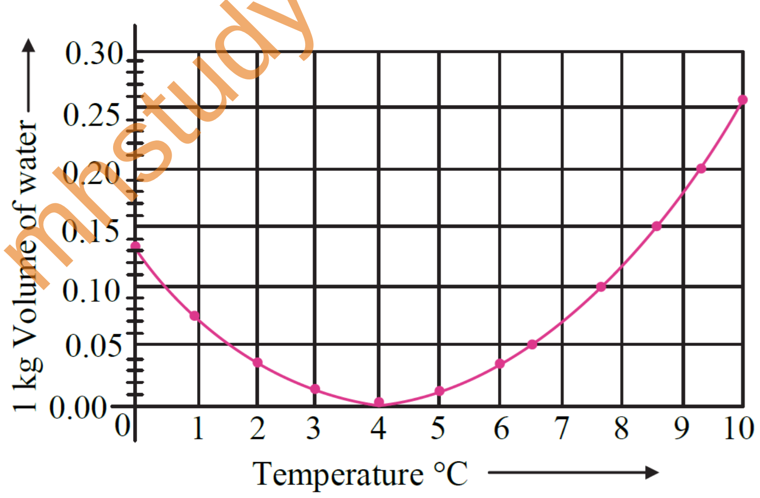
Previous Year Questions

1. **Explain the following temperature Vs time graph: Exercise Question 5**



2. **How much heat energy is necessary to raise the temperature of 5 kg of water from 20°C to 100°C? – Numericals at the end**

3. **Observe the given graph and answer the following questions:**



   a. **Name the process represented in the figure.**

   Anomalous behaviour of water

   b. **At what temperature does this process take place?**

   Between 0°C and 4°C

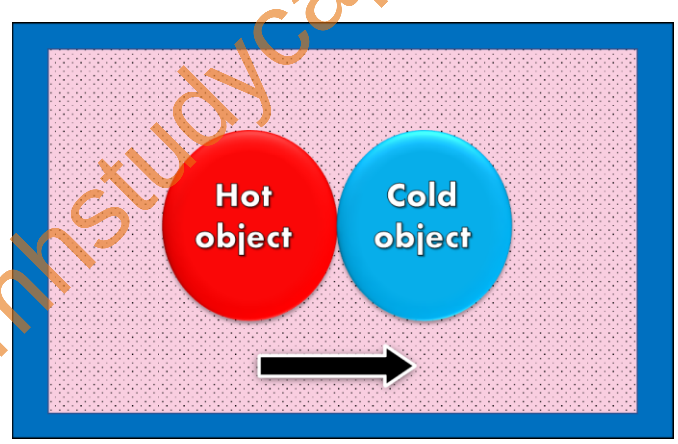
4. Define specific heat capacity. Write its S.I. unit.

i. The quantity of heat energy required to raise a temperature of unit mass of a substance through 1°C is called specific heat capacity of a substance.

ii. Specific heat capacity is denoted by letter 'c'. In CGS, it is measured in cal/g°C, while in MKS, it is measured in J/kg°C.

5. A copper sphere of 100 g mass is heated to raise its temperature to 100°C and is released in water of mass 195 g and temperature 20°C in a copper calorimeter. If the mass of the calorimeter is 50 g, what will be the maximum temperature of water? (Specific heat of copper = 0.1 cal/g°C) – Numericals at the end

6. Answer the following questions with the help of following diagram.



(a) Heat is transferred from where to where?

Heat is transferred from hot object to cold object.

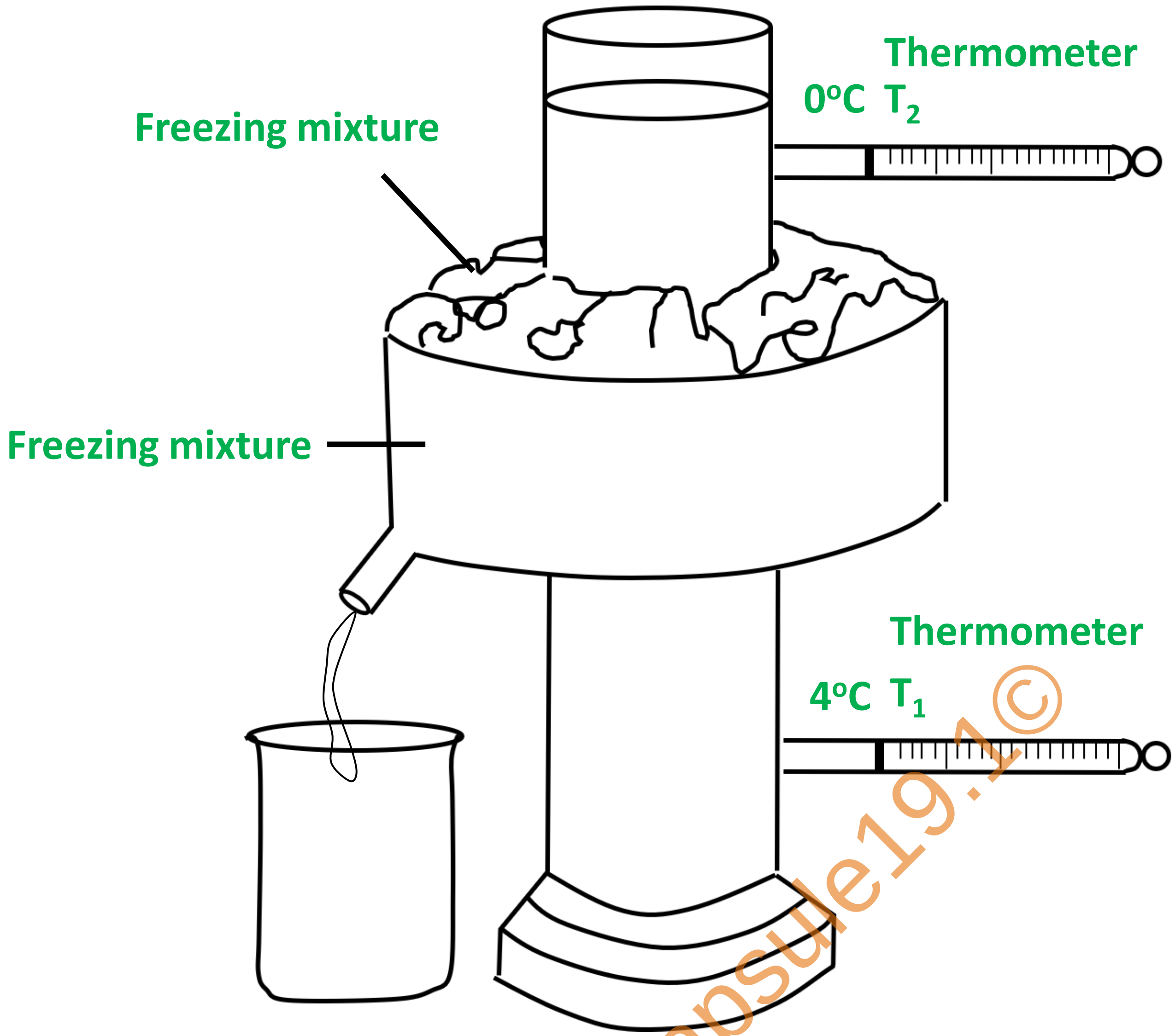
(b) Which principle do we learn about from this process?

The principle of heat exchange.

(c) Which property of the substance is measured using this principle?

Specific heat capacity of a substance is measured by this principle.

7. **Draw a neat labelled diagram of Hope's Apparatus.**



8. **Read the following paragraph and answer the questions based on it: - Exercise Question 8**

9. **How do we feel about air in each of the following conditions?**

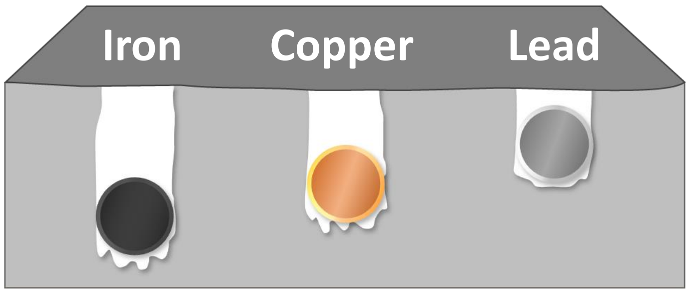
   (a) Relative humidity is more than 60%.

   We feel the air is humid

   (b) Relative humidity is less than 60%.

   We feel the air is dry

10. Observe the following figure and answer the questions:



(a) Which element has maximum specific heat capacity? Justify.

Maximum specific heat capacity is seen in iron. The iron sphere absorbs more heat, which results in more heat transmission in the wax, so the sphere goes the deepest into the wax.

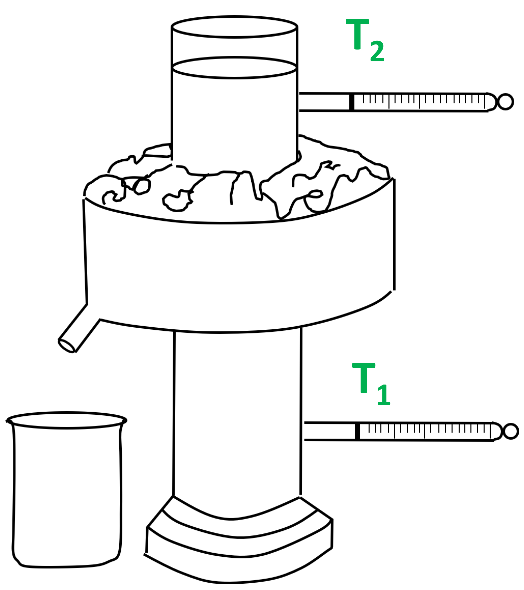
(b) Which element has minimum specific heat capacity? Justify.

Minimum specific heat capacity is seen in lead. The lead sphere penetrates the wax the least because it absorbs less heat and transmits less heat to the wax

(c) Define specific heat of substance.

The quantity of heat energy required to raise a temperature of unit mass of a substance through 1°C is called specific heat capacity of a substance.

11. Observe the given diagram and answer the following questions:



(a) What is the name of the given apparatus?

Hope's Apparatus

(b) Which phenomenon is studied with the help of this apparatus?

Anomalous behaviour of water

(c) What are the final temperatures in thermometers T1 and T2?

Final temperature in T_1 and T_2 are 0^oC.

(d) At what temperature, the density of water is maximum?

At 4^oC

(e) Give one example of the above phenomenon in nature.

Freezing of top layer of lake